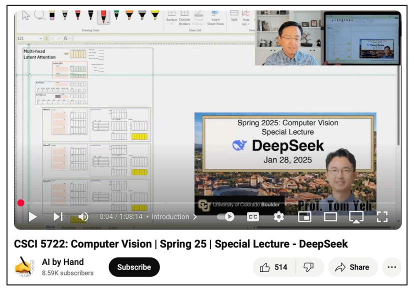

https://www.youtube.com/watch?v=idF6TiTGYsE

Multi-head Latent Attention

deepseek

Input

t	1	2	3	4	5	6
	1	0	1	1	0	0
	0	0	-1	0	0	-1
	1	0	1	0	0	0
	1	0	1	0	-1	0
	1	0	1	0	0	0
	↓	↓	↓	↓	↓	↓

Model Size	5
Context length	6
# of heads	3
per-head dimensions	4

Compression Dimensions

query	3
key/value	2

Latent QKV

W^DQ

-1	-1	0	0	1
1	0	1	0	0
2	-1	0	1	1

C_t^Q

0	0	1	-1	0	1
2	0	2	1	0	0
4	0	5	2	-1	1

W^DKV

1	0	-1	-1	0
1	0	1	0	0

C_t^KV

-1	0	-1	1	1	0
2	0	2	1	0	0

RoPE (Key)

W^KR

0	0	0	1	1
1	0	1	1	0

2	0	2	0	-1	0
3	0	3	1	-1	0

R1	10	R2	20	R3	30	R4	40	R5	50	R6	60
-0.8	0.54	0.41	-0.9	0.15	0.99	-0.7	-0.7	0.96	0.26	-1	0.3
-0.5	-0.8	0.91	0.41	-1	0.15	0.75	-0.7	-0.3	0.96	-0.3	-1

K^R

-0	0	3.27	-0.7	-1.2	0
-3.6	0	-1.5	-0.7	-0.7	0

RoPE (Query)

W^QR

		1	0.5	1
		-1	1	0.5

5	0	7	1.5	-1	2
4	0	3.5	3	-0.5	-0.5

R1		R2		R3		R4		R5		R6	
-0.8	0.54	0.41	-0.9	0.15	0.99	-0.7	-0.7	0.96	0.26	-1	0.3
-0.5	-0.8	0.91	0.41	-1	0.15	0.75	-0.7	-0.3	0.96	-0.3	-1

Q^R

-2	0	4.54	-3.2	-1.1	-2.1
-6.1	0	-6.4	-0.9	-0.2	-0.1

Head 1

Linear

Wq → Q^C

1	1	-1	1	1	→	6	0	6	4	-1	0
1	0	1	1	1	→	6	0	8	2	-1	2
1	0	0	1	0	→	2	0	2	1	0	0
1	1	0	1	0	→	2	0	2	1	0	0

Wk → K^C

1	1	-1	1	1	→	1	0	1	2	1	0
0	0	0	0	0	→	0	0	0	0	0	0
0	1	0	0	1	→	2	0	2	1	0	0
1	1	0	1	0	→	-1	0	-1	1	1	0

Wv → V^C

1	1	-1	1	1	→	1	0	1	2	1	0
0	0	1	0	1	→	2	0	2	1	0	0
1	1	0	1	0	→	-1	0	-1	1	1	0
0	1	0	1	0	→	-1	0	-1	1	1	0

Scaled Dot-Product

Q^R	-2	0	4.54	-3.2	-1.1	-2.1
	-6.1	0	-6.4	-0.9	-0.2	-0.1
Q^C	6	0	6	4	-1	0
	6	0	8	2	-1	2
	2	0	2	1	0	0
	2	0	2	1	0	0

K^R, K^C → K^T * Q / sqrt(dk)

	K^R		K^C					↓	↓	↓	↓	↓	↓
1	-0	-3.6	1	0	2	-1	→	30	0	30.8	8.33	-0.2	0.58
2	0	0	0	0	0	0	→	0	0	0	0	0	0
3	3.27	-1.5	1	0	2	-1	→	10.6	0	32.5	-4.3	-4.3	-6.5
4	-0.7	-0.7	2	0	1	1	→	21.6	0	16.9	13	-1	1.62
5	-1.2	-0.7	1	0	0	1	→	14.7	0	6.91	9.59	0.5	2.62
6	0	0	0	0	0	0	→	0	0	0	0	0	0

Softmax

	1	2	3	4	5	6
1	1	0.17	0.15	0.01	0.18	0.08
2	0	0.17	0	0	0.21	0.04
3	0	0.17	0.85	0	0	0
4	0	0.17	0	0.96	0.07	0.22
5	0	0.17	0	0.03	0.34	0.61
6	0	0.17	0	0	0.21	0.04

V^C → V'

							↓	↓	↓	↓	↓	↓
1	0	1	2	1	0	→	1	0.83	1	1.96	0.66	1.14
2	0	2	1	0	0	→	2	0.83	2	0.98	0.43	0.38
-1	0	-1	1	1	0	→	-1	0	-1	0.98	0.23	0.75
-1	0	-1	1	1	0	→	-1	0	-1	0.98	0.23	0.75

Head 2

Linear

Wq → Q^C

1	0	1	0	0	→	2	0	2	1	0	0
0	1	0	0	1	→	1	0	0	0	0	-1
0	1	0	0	0	→	0	0	-1	0	0	-1
-1	-1	0	0	0	→	-1	0	0	-1	0	1

Wk → K^C

1	0	0	0	0	→	1	0	1	1	0	0
0	0	0	1	0	→	1	0	1	0	-1	0
1	0	1	0	0	→	2	0	2	1	0	0
1	0	1	0	1	→	3	0	3	1	0	0

Scaled Dot-Product

Q^R	-2	0	4.54	-3.2	-1.1	-2.1
	-6.1	0	-6.4	-0.9	-0.2	-0.1
Q^C	2	0	2	1	0	0
	1	0	0	0	0	-1
	0	0	-1	0	0	-1
	-1	0	0	-1	0	1

K^R, K^C → K^T * Q / sqrt(dk)

	K^R		K^C					↓	↓	↓	↓	↓	↓
1	-0	-3.6	1	1	2	3	→	22	0	22.8	1.33	0.84	0.58
2	0	0	0	0	0	0	→	0	0	0	0	0	0
3	3.27	-1.5	1	1	2	3	→	2.59	0	24.5	-11	-3.3	-6.5

Softmax

	1	2	3	4	5	6
1	1	0.17	0.15	0.03	0.2	0.04
2	0	0.17	0	0.01	0.09	0.02
3	0	0.17	0.85	0	0	0

Wv						V^C					
1	0	0	-1	0		0	0	0	1	1	0
0	1	1	0	0		1	0	0	0	0	-1
1	1	0	1	0		2	0	1	1	-1	-1
1	-1	1	0	0		2	0	3	1	0	1

	K^R		K^C													Softmax					
4	-0.7	-0.7	1	0	1	1	6.56	0	1.87	3	0.96	1.62	4	0	0.17	0	0.16	0.23	0.11
5	-1.2	-0.7	0	-1	0	0	5.75	0	-1.1	4.59	1.5	3.62	5	0	0.17	0	0.79	0.39	0.81
6	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0.17	0	0.01	0.09	0.02

V^C						V'					
0	0	0	1	1	0	0	0.33	0	0.95	0.62	0.92
1	0	0	0	0	-1	1	0	0.15	0.02	0.12	0.02
2	0	1	1	-1	-1	2	0.33	1.15	-0.6	0.16	-0.6
2	0	3	1	0	1	2	1.17	2.85	0.23	0.73	0.21

Head 3

Linear

Wq						Q^C					
0	0	1	0	0		1	0	1	0	0	0
1	0	0	1	0		2	0	2	1	-1	0
0	1	0	1	0		1	0	0	0	-1	-1
1	0	1	0	0		2	0	2	1	0	0

Wk						K^C					
0	1	0	0	1		1	0	0	0	0	-1
0	0	1	0	1		2	0	2	0	0	0
0	0	1	1	0		2	0	2	0	-1	0
1	0	1	0	1		3	0	3	1	0	0

Wv						V^C					
1	1	0	-1	0		0	0	-1	1	1	-1
0	0	1	1	0		2	0	2	0	-1	0
1	0	0	1	1		3	0	3	1	-1	0
1	-1	1	0	0		2	0	3	1	0	1

Scaled Dot-Product

Q^R	-2	0	4.54	-3.2	-1.1	-2.1
	-6.1	0	-6.4	-0.9	-0.2	-0.1
Q^C	1	0	1	0	0	0
	2	0	2	1	-1	0
	1	0	0	0	-1	-1
	2	0	2	1	0	0

	K^R		K^C				K^T * Q / sqrt(dk)						Softmax	1	2	3	4	5	6
1	-0	-3.6	1	2	2	3	35	0	33.8	8.33	-3.2	-1.4	1	1	0.17	0.33	0.96	0	0.01
2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0.17	0	0	0.06	0.02
3	3.27	-1.5	0	2	2	3	14.6	0	34.5	-4.3	-7.3	-8.5	3	0	0.17	0.67	0	0	0
4	-0.7	-0.7	0	0	0	1	7.56	0	2.87	4	0.96	1.62	4	0	0.17	0	0.01	0.16	0.11
5	-1.2	-0.7	0	0	-1	0	5.75	0	-1.1	4.59	2.5	3.62	5	0	0.17	0	0.02	0.72	0.84
6	0	0	-1	0	0	0	-1	0	-1	0	0	0	6	0	0.17	0	0	0.06	0.02

V^C						V'					
0	0	-1	1	1	-1	-0	0	-0.7	0.04	0.82	0.93
2	0	2	0	-1	0	2	0.5	2	1.91	-0.7	-0.8
3	0	3	1	-1	0	3	1	3	2.88	-0.6	-0.7
2	0	3	1	0	1	2	1.17	2.67	1.94	0.22	0.15

Concat

Head	V'					
1	1	0.83	1	1.96	0.66	1.14
	2	0.83	2	0.98	0.43	0.38
	-1	0	-1	0.98	0.23	0.75
	-1	0	-1	0.98	0.23	0.75
2	0	0.33	0	0.95	0.62	0.92
	1	0	0.15	0.02	0.12	0.02
	2	0.33	1.15	-0.6	0.16	-0.6
	2	1.17	2.85	0.23	0.73	0.21
3	-0	0	-0.7	0.04	0.82	0.93
	2	0.5	2	1.91	-0.7	-0.8
	3	1	3	2.88	-0.6	-0.7
	2	1.17	2.67	1.94	0.22	0.15

Output Projection

Wo												Output					
1	1	0	-1	0	1	0	1	0	1	0	1	11	4.5	11.7	6.05	1.21	0.31
1	0	-1	0	1	0	1	0	1	0	2	0	10	3.5	8.48	7.15	0.91	0.17
0	0	1	0	-1	0	1	0	0	0	1	-1	2	-0.2	0.48	0.39	-1	-1.7
0	1	0	0	0	-1	-1	0	0	1	1	0	4	2	5.7	6.32	-1.1	-0.5
1	-1	0	1	0	1	0	0	0	1	0	0	1	0.5	0.15	3.89	-0.1	0.7

Output

Mixture of Experts

Input

t=	1	2	3	4	5	6
	11	4.5	11.7	6.05	1.21	0.31
	10	3.5	8.48	7.15	0.91	0.17
	2	-0.2	0.48	0.39	-1	-1.7
	4	2	5.7	6.32	-1.1	-0.5
	1	0.5	0.15	3.89	-0.1	0.7
	1	1	1	1	1	1

Intermedia dimensions 7

# of Experts 4

# of shared 1

Router

Shared	E1											
Routed	E2	1	0	0	1	0	15	6.5	17.4	12.4	0.09	-0.2
	E3	0	1	1	0	2	14	4.33	9.26	15.3	-0.4	-0.1
	E4	1	2	1	0	1	34	11.8	29.3	24.6	1.88	-0.3

Affinity Scores

Sigmoid					
1	1	1	1	1	1
1	1	1	1	0.52	0.45
1	0.99	1	1	0.41	0.48
1	1	1	1	0.87	0.42

Rank	Sort by Columns					
1	1	1	1	1	1	1
2	1	1	1	1	0.87	0.48
3	1	1	1	1	0.52	0.45
4	1	0.99	1	1	0.41	0.42

Gates

Top3						
E1	1	1	1	1	1	1
E2	1	1	1	0	0.52	0.45
E3	0	0	0	1	0	0.48
E4	1	1	1	1	0.87	0

Expert 1

Linear

W					b						
1	0	-1	-1	0	1	6	3.67	6.5	0.34	4.34	3.5
1	0	1	0	0	2	15	6.33	14.2	8.44	2.2	0.65
1	1	0	0	0	1	22	9	21.2	14.2	3.12	1.48
1	1	1	0	1	2	26	10.3	22.8	19.5	2.97	1.52
-1	-1	1	0	1	1	-17	-6.7	-19	-7.9	-2.3	-0.4
0	0	1	0	0	2	4	1.83	2.48	2.39	0.99	0.34
1	-1	0	1	1	-1	5	2.5	8.04	8.11	-2	-0.7

ReLU						
	6	3.67	6.5	0.34	4.34	3.5

Expert 2

Linear

W					b						
1	0	1	-1	0	1	10	3.33	7.46	1.12	2.32	0.17
1	1	1	0	0	2	25	9.83	22.6	15.6	3.11	0.82
1	0	0	0	0	1	12	5.5	12.7	7.05	2.21	1.31
1	0	1	0	1	2	16	6.83	14.3	12.3	2.06	1.35
-1	0	1	1	1	1	-3	-1.2	-4.3	5.55	-2.5	-0.8
0	0	1	0	0	2	4	1.83	2.48	2.39	0.99	0.34
1	-1	0	1	0	-1	4	2	7.89	4.22	-1.8	-1.4

ReLU						
	10	3.33	7.46	1.12	2.32	0.17

Expert 3

Linear

W					b						
1	0	-1	-1	0	1	6	3.67	6.5	0.34	4.34	3.5
1	0	1	0	0	2	15	6.33	14.2	8.44	2.2	0.65
1	1	0	0	0	1	22	9	21.2	14.2	3.12	1.48
1	0	1	0	1	0	14	4.83	12.3	10.3	0.06	-0.7
-1	-1	1	0	1	1	-17	-6.7	-19	-7.9	-2.3	-0.4
0	0	1	0	0	0	2	-0.2	0.48	0.39	-1	-1.7
1	-1	0	1	0	-1	4	2	7.89	4.22	-1.8	-1.4

ReLU						
	6	3.67	6.5	0.34	4.34	3.5

Expert 4

Linear

W					b						
1	0	-1	-1	0	1	6	3.67	6.5	0.34	4.34	3.5
0	0	1	0	0	2	4	1.83	2.48	2.39	0.99	0.34
1	1	0	0	0	1	22	9	21.2	14.2	3.12	1.48
1	0	1	0	1	2	16	6.83	14.3	12.3	2.06	1.35
-1	-1	1	0	1	1	-17	-6.7	-19	-7.9	-2.3	-0.4
0	0	0	0	0	2	2	2	2	2	2	2
1	-1	0	0	1	-1	1	0.5	2.35	1.79	-0.8	-0.2

ReLU						
	6	3.67	6.5	0.34	4.34	3.5

15	6.33	14.2	8.44	2.2	0.65
22	9	21.2	14.2	3.12	1.48
26	10.3	22.8	19.5	2.97	1.52
0	0	0	0	0	0
4	1.83	2.48	2.39	0.99	0.34
5	2.5	8.04	8.11	0	0
1	1	1	1	1	1

Linear

W							b
0	1	0	-1	-1	0	0	1
0	1	0	1	0	0	0	2
-1	1	1	0	0	0	1	1
-1	1	1	1	0	1	0	2
1	-1	-1	1	0	1	0	1

-10	-3	-7.6	-10	0.23	0.13
43	18.7	38.9	29.9	7.17	4.17
37	15.2	37.8	31.4	1.98	-0.4
63	25.8	56.1	46.2	6.95	2.5
0	1.5	-2.5	0.57	3.99	4.22

ReLU

0	0	0	0	0.23	0.13
43	18.7	38.9	29.9	7.17	4.17
37	15.2	37.8	31.4	1.98	0
63	25.8	56.1	46.2	6.95	2.5
0	1.5	0	0.57	3.99	4.22

Gates

E1	1	1	1	1	1	1

0	0	0	0	0.23	0.13
43	18.7	38.9	29.9	7.17	4.17
37	15.2	37.8	31.4	1.98	0
63	25.8	56.1	46.2	6.95	2.5
0	1.5	0	0.57	3.99	4.22

25	9.83	22.6	15.6	3.11	0.82
12	5.5	12.7	7.05	2.21	1.31
16	6.83	14.3	12.3	2.06	1.35
0	0	0	5.55	0	0
4	1.83	2.48	2.39	0.99	0.34
4	2	7.89	4.22	0	0
1	1	1	1	1	1

Linear

W							b
0	1	0	-1	-1	0	0	0
0	0	0	1	1	1	0	2
-1	1	1	1	0	0	1	1
-1	0	1	1	1	1	0	2
1	0	-1	1	0	1	0	1

9	3	8.33	-2.3	1.05	-0.5
22	10.7	18.8	22.3	5.06	3.69
48	21.8	51	39.1	6.06	4.31
24	12.8	24	28.2	4.94	4.83
19	7.5	12.6	9.8	4.17	1.55

ReLU

9	3	8.33	0	1.05	0
22	10.7	18.8	22.3	5.06	3.69
48	21.8	51	39.1	6.06	4.31
24	12.8	24	28.2	4.94	4.83
19	7.5	12.6	9.8	4.17	1.55

Gates

E2	1	1	1	0	0.52	0.45

9	3	8.33	0	0.55	0
22	10.7	18.8	0	2.64	1.65
48	21.8	51	0	3.16	1.93
24	12.8	24	0	2.58	2.16
19	7.49	12.6	0	2.18	0.69

15	6.33	14.2	8.44	2.2	0.65
22	9	21.2	14.2	3.12	1.48
14	4.83	12.3	10.3	0.06	0
0	0	0	0	0	0
2	0	0.48	0.39	0	0
4	2	7.89	4.22	0	0
1	1	1	1	1	1

Linear

W							b
0	1	0	-1	-1	0	0	1
0	0	0	1	0	1	0	2
-1	1	1	0	0	1	1	0
-1	0	1	0	0	1	0	2
1	-1	-1	1	0	1	0	1

2	2.5	2.85	-0.9	3.14	1.65
18	6.83	14.8	12.7	2.06	2
37	13.7	37.2	26.9	0.98	-1.4
20	7.33	17.1	16.3	0.78	-0
-14	-5.8	-15	-11	0.08	2.36

ReLU

2	2.5	2.85	0	3.14	1.65
18	6.83	14.8	12.7	2.06	2
37	13.7	37.2	26.9	0.98	0
20	7.33	17.1	16.3	0.78	0
0	0	0	0	0.08	2.36

Gates

E3	0	0	0	1	0	0.48

0	0	0	0	0	0.79
0	0	0	12.7	0	0.95
0	0	0	26.9	0	0
0	0	0	16.3	0	0
0	0	0	0	0	1.13

4	1.83	2.48	2.39	0.99	0.34
22	9	21.2	14.2	3.12	1.48
16	6.83	14.3	12.3	2.06	1.35
0	0	0	0	0	0
2	2	2	2	2	2
1	0.5	2.35	1.79	0	0
1	1	1	1	1	1

Linear

W							b
0	1	0	-1	-1	1	0	1
0	1	0	0	0	0	0	0
-1	1	1	0	0	0	1	1
0	0	1	1	0	1	0	2
1	-1	-1	1	0	1	0	1

-9	-2	-8.8	-6.9	1.93	1.99
4	1.83	2.48	2.39	0.99	0.34
22	8.67	20.5	19	0.77	-0.7
42	19.8	39.5	30.5	9.18	6.83
-1	2.67	0.17	-0.9	5.29	6.02

ReLU

0	0	0	0	1.93	1.99
4	1.83	2.48	2.39	0.99	0.34
22	8.67	20.5	19	0.77	0
42	19.8	39.5	30.5	9.18	6.83
0	2.67	0.17	0	5.29	6.02

Gates

E4	1	1	1	1	0.87	0

0	0	0	0	1.67	0
4	1.83	2.48	2.39	0.86	0
22	8.67	20.5	19	0.67	0
42	19.8	39.5	30.5	7.97	0
0	2.67	0.17	0	4.59	0

Mixture

20	7.5	20	6.05	3.66	1.23
79	34.7	68.7	52.2	11.6	6.95
109	45.5	110	77.8	4.8	0.27
133	60.5	125	99.3	16.4	4.13
20	12.2	12.9	4.46	10.6	6.74

Output